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K J Somaiya Institute of Technology
(Formerly known as K J Somaiya Institute of Engineering and Information Technology)
An Autonomous Institute Permanently Affiliated to University of Mumbai.

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T.Y. Mini Project Presentation

"Deep Learning Time-Series Forecasting of River Streamflow in Drought-Prone Basins(Krishna-Cauvery) under El Nino"

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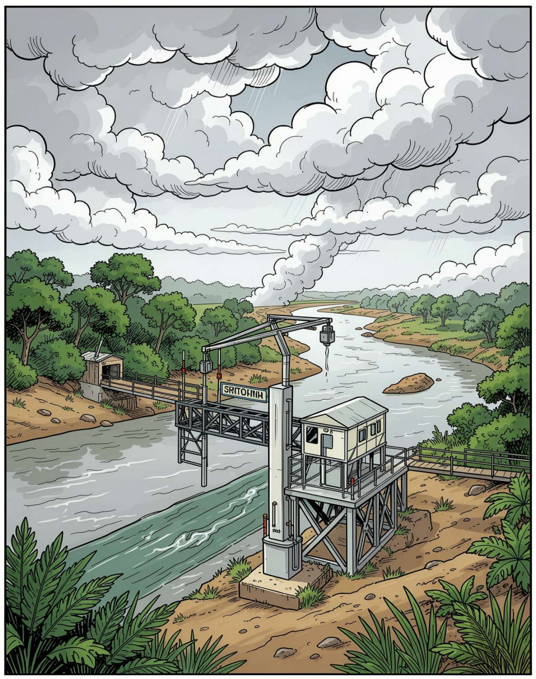
Introduction



What is Streamflow?
-> Water volume flowing through a river per unit time

Target Basins
-> Krishna & Cauvery: drought-prone, El Niño-influenced regions

Our Approach
-> Temporal Convolutional Networks (TCN) for time-series forecasting

Impact
-> Supports water security, drought early-warning & inter-state planning

Literature Survey

Traditional Models
Physics-based, require extensive manual calibration

ML Approaches
Capture non-linear rainfall-flow relationships effectively

TCN Advantages
Parallel processing, stable gradients, controllable receptive field

RNN / LSTM Limits
Sequential processing, slow training, gradient instability

El Niño Linkage
Statistical but significant correlation with monsoon & streamflow

Project Objective

Model Design

TCN-based deep learning model for streamflow forecasting

Input Variables

Rainfall, temperature, streamflow & El Niño index

Forecast Window

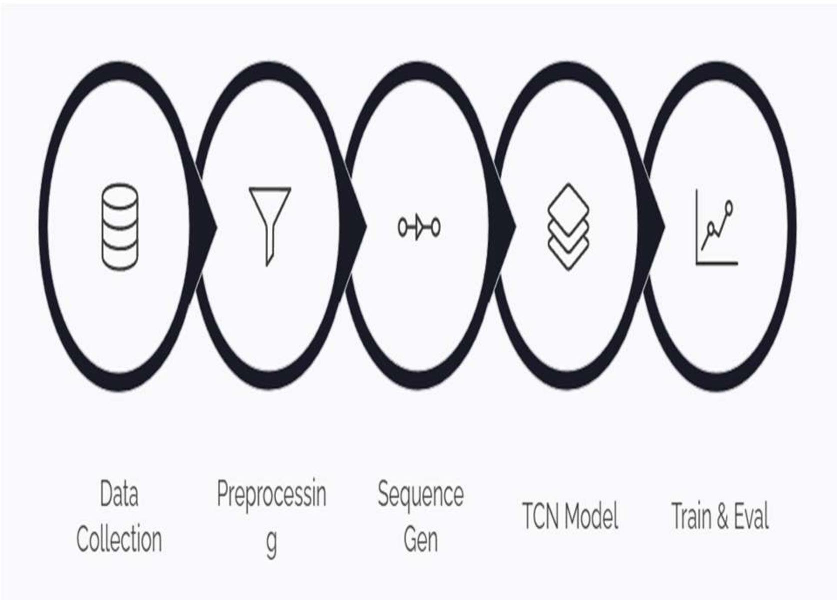
7–30 day future streamflow at river gauge stations

Evaluation Metrics

RMSE, MAE, NSE, and R^2 for performance quantification

Focus on reliable predictions specifically during **El Niño-driven drought years** — the hardest forecasting scenario.

Implementation



01

Data Sources
CWC streamflow, IMD rainfall/temperature, NOAA El Niño index

02

Preprocessing
Handle missing values, normalize features, chronological split

03

Sliding Window
30-day input sequences → 7-day forecast horizon

04

TCN Architecture
Causal, dilated convolutions with residual connections

05

Training
MSE loss, Adam optimizer, validation monitoring for overfitting

Expected Result

m^3/s	4	7-30
Forecast Output Numeric streamflow for Krishna & Cauvery basins	**Key Metric** RMSE, MAE, NSE & R^2 quantify prediction accuracy	**Day Horizon** Directional drought warnings via early low-flow detection

Key Findings Anticipated

- Accurate forecasts under normal & moderate El Niño conditions
- Lower accuracy during extreme drought years (data scarcity)
- Reusable pipeline adaptable to other Indian river basins

Application

Reservoir & Irrigation
Optimise release scheduling & irrigation planning

Drought Early-Warning
Proactive alerts for disaster management authorities

Inter-State Water Sharing
Data-driven decision support for water-sharing disputes

Hydropower Scheduling
Forecast generation capacity & optimise turbine scheduling

Climate-Hydrology Research
Advance understanding of Indian river basin dynamics

Future Scope

Multi-Basin Forecasting

Spatially distributed models across multiple basins

Hybrid TCN + Physics

Combine data-driven & physics-based hydrological models

Uncertainty Estimation

Confidence intervals for probabilistic drought forecasts

Additional Climate Indices

Incorporate Indian Ocean Dipole & other drivers

Conclusion

TCN vs LSTM

Faster, more stable alternative for streamflow forecasting

Temporal Patterns

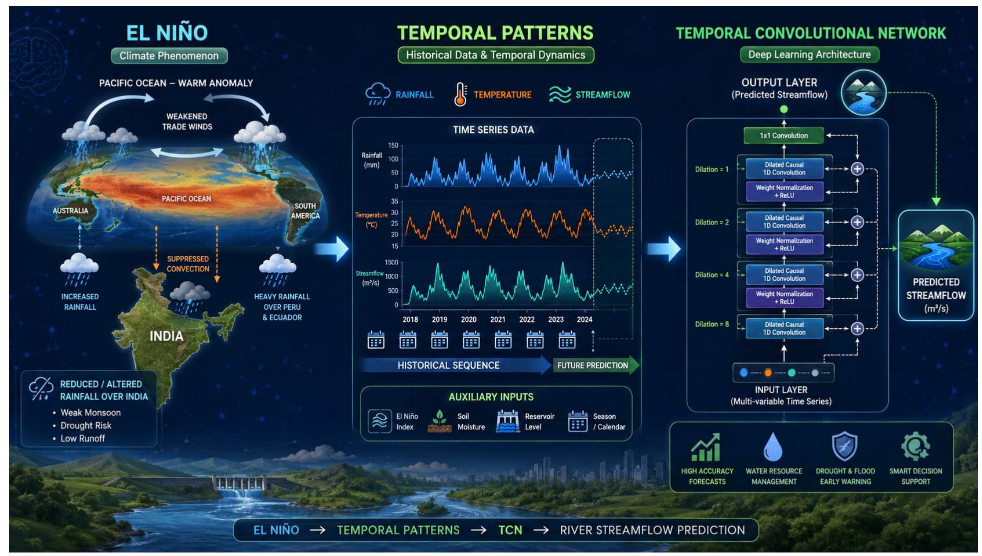
Causal & dilated convolutions capture short & long-term rainfall

El Niño Integration

Index as input enables effective drought-period learning

Complementary Tool

Data-driven model complements traditional hydrological methods



Ultimately supports **water security & climate adaptation** across India's drought-prone river basins.

References

01 **Central Water Commission (CWC)**
Streamflow gauge data — primary hydrological dataset

02 **India Meteorological Department (IMD)**
Rainfall & temperature datasets for model inputs

03 **NOAA Climate Prediction Center**
Oceanic Niño Index (ONI) — El Niño climate driver

04 **Bai et al. (2018)**
Temporal Convolutional Networks for sequence modelling

Time Plan

1. **Phase 1 — Week 1–2**
   Data collection & preprocessing
2. **Phase 2 — Week 3–4**
   TCN model architecture design & implementation
3. **Phase 3 — Week 5–6**
   Model training, validation & hyperparameter tuning
4. **Phase 4 — Week 7**
   Testing, evaluation & drought analysis
5. **Phase 5 — Week 8**
   Documentation & presentation preparation